

TWO CIRCLES MAKE AN ELLIPSE: THE CONE'S PARABOLAS AND HYPERBOLA AS EPICYCLE INVARIANTS OF THE LEVI-CIVITA POSITION SPINOR

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ABSTRACT. [4] writes the fictitious-time position spinor as $u(s) = \sqrt{y} \cos \theta + i\sqrt{x} \sin \theta$, $\theta = \omega s + \phi_0$, and shows it traces an ellipse, not a circle, in its own complex plane. We show this ellipse decomposes exactly into two counter-rotating circular components, $u = \frac{1}{2}(A_+ + A_-)$, $A_{\pm} = R_{\pm} e^{\pm i\theta}$, the classical epicycle (Ptolemaic) decomposition of an ellipse, with $R_+ = \sqrt{x} + \sqrt{y}$, $R_- = \sqrt{y} - \sqrt{x}$. We show every coordinate of the cone is an exact function of these two radii: $Y = (R_+^2 - R_-^2)/4$, $T = (R_+^2 + R_-^2)/4$, and $X = -\frac{1}{2}R_+R_-$ (via the constant product $A_+A_- = -2X$). This gives all four of the cone's own curve families a uniform reading. The row and column parabolas of [1], already connected in earlier work to the angular momentum $L_u = \omega Y$ of the spinor's own rotation, are subsumed here. Two further parabola families, obtained by cutting the cone parallel to the *other* pair of generators (fixing $T \mp Y$ rather than a linear combination of x, y), turn out to fix a single epicycle radius: $T - Y = k \iff R_-^2 = 2k$ and $T + Y = k \iff R_+^2 = 2k$. And the hyperbola of [1], the orbit of the boost generator B_X , induces on (R_+, R_-) an exact boost at *half* the rapidity — the direct hyperbolic-generator counterpart to the half-angle rotation of [4]'s own Proposition 4.1 — under which the identity $R_+^2 - R_-^2 = 4Y$ reproduces the cone's own hyperbola one level down, with the same shape identified in the companion note [5] as the Ellis–Bronnikov wormhole embedding, now with throat $2\sqrt{Y}$. Every claim below is verified symbolically and, where relevant, numerically.

1. SETUP

Recall from [4]: for a fixed unperturbed Kepler orbit with $x = T + X = r_a$, $y = T - X = r_p$, the fictitious-time position spinor is

$$(1) \quad u(s) = \sqrt{y} \cos \theta + i\sqrt{x} \sin \theta, \quad \theta = \omega s + \phi_0, \quad \omega = \frac{1}{2}\sqrt{\mu/T},$$

and u traces an ellipse, not a circle, of semi-axes $\sqrt{y}, \sqrt{x}$ in its own complex plane — $|u(s)|^2$ ranges between $x = r_a$ and $y = r_p$, not constant.

2. THE EPICYCLE DECOMPOSITION

Theorem 2.1 (Two circles). *Define $A_{\pm}(s) := u(s) \mp i u'(s)/\omega$. Then, exactly,*

$$(2) \quad A_+(s) = (\sqrt{x} + \sqrt{y}) e^{i\theta}, \quad A_-(s) = (\sqrt{y} - \sqrt{x}) e^{-i\theta},$$

so that $u(s) = \frac{1}{2}(A_+(s) + A_-(s))$: the position spinor is the exact average of two circular motions of constant radii $R_+ := \sqrt{x} + \sqrt{y}$ and $R_- := \sqrt{y} - \sqrt{x}$, rotating at $+\omega$ and $-\omega$ respectively.

Proof. From (1), $u'(s) = \omega(-\sqrt{y} \sin \theta + i\sqrt{x} \cos \theta)$. Direct expansion gives

$$u - \frac{iu'}{\omega} = (\sqrt{x} + \sqrt{y})(\cos \theta + i \sin \theta), \quad u + \frac{iu'}{\omega} = (\sqrt{y} - \sqrt{x})(\cos \theta - i \sin \theta),$$

which are (2); averaging the two recovers u . Verified symbolically. $\square$

This is the classical epicycle (two-counter-rotating-circles) decomposition of an ellipse, applied here to the fictitious-time position spinor itself rather than to the physical orbit — and it repairs, with an exact statement, the informal picture of [4]’s earlier draft: it is not u but each of A_+, A_- individually that traces a genuine circle at rate ω .

3. THE FULL COORDINATE DICTIONARY

Proposition 3.1 ((X, Y, T) from the epicycle radii).

$$(3) \quad R_+^2 = 2(T + Y), \quad R_-^2 = 2(T - Y), \quad A_+ A_- = -2X.$$

Equivalently, $T = \frac{1}{4}(R_+^2 + R_-^2)$, $Y = \frac{1}{4}(R_+^2 - R_-^2)$, and $X = -\frac{1}{2}R_+ R_-$.

Proof. $R_+^2 = x + y + 2\sqrt{xy} = 2T + 2Y$; $R_-^2 = x + y - 2\sqrt{xy} = 2T - 2Y$, using $Y = \sqrt{xy}$, $T = (x + y)/2$. For the product: $A_+ A_- = R_+ R_- e^{i\theta} e^{-i\theta} = R_+ R_- = (\sqrt{x} + \sqrt{y})(\sqrt{y} - \sqrt{x}) = y - x = -2X$, a genuinely s -independent (conserved) quantity, verified symbolically. $\square$

So every one of the cone’s own coordinates is recovered from the two epicycle radii of the fictitious-time motion, exactly, with no approximation.

4. THE ROW/COLUMN FAMILY: ANGULAR MOMENTUM

The row and column parabolas of [1] fix x (resp. y) and vary the other, giving $Y^2 = xy = cW$ for the free coordinate W . This family was already connected, independently of the present note, to the spinor’s own angular momentum $L_u := \text{Im}(\bar{u} u')$:

Proposition 4.1. $L_u = \omega Y$, hence $h = 2\omega Y = 2L_u$, where $h = Y\sqrt{\mu/T}$ is the physical specific angular momentum.

Proof. Direct computation from (1): $L_u = u_1 u'_2 - u_2 u'_1 = \omega \sqrt{xy} = \omega Y$, verified symbolically. The relation $h = 2\omega Y$ follows from $\omega = \frac{1}{2}\sqrt{\mu/T}$ and $h = Y\sqrt{\mu/T}$ by substitution. $\square$

So the row/column family, revolved, is the locus “the geometric mean of the epicycle radii’s squares” — more precisely, Y itself is L_u/ω , the spinor’s angular momentum divided by its frequency.

5. THE OTHER TWO PARABOLAS: THE EPICYCLE RADII THEMSELVES

The row and column planes are parallel to the cone’s generators $(\pm 1, 0, 1)$ (zero Y -component). The cone’s other pair of generators, $(0, \pm 1, 1)$, gives a second, independent parabola family: planes $T \mp Y = k$.

Proposition 5.1. $T - Y = k \iff (\sqrt{x} - \sqrt{y})^2 = 2k \iff R_-^2 = 2k$, and $T + Y = k \iff R_+^2 = 2k$.

Proof. $T - Y = \frac{1}{2}(x + y) - \sqrt{xy} = \frac{1}{2}(\sqrt{x} - \sqrt{y})^2$, and $(\sqrt{x} - \sqrt{y})^2 = R_-^2$ (since $R_- = \sqrt{y} - \sqrt{x}$ and squaring removes the sign); the second identity is the mirror computation with $T + Y = \frac{1}{2}(\sqrt{x} + \sqrt{y})^2 = \frac{1}{2}R_+^2$. $\square$

So these two parabola families — genuinely absent from [1, 2], which classify only linear cuts in (x, y) itself, not in $(\sqrt{x}, \sqrt{y})$ — are exactly the level sets *of the two epicycle radii individually*: $T - Y = k$ holds the clockwise epicycle's radius fixed at $\sqrt{2k}$, and $T + Y = k$ holds the counterclockwise one fixed. Revolving either (about the axis along which the curve is symmetric in X) gives a paraboloid of the same algebraic type as the row family, but now reading a single dynamical invariant of the spinor's own motion rather than a product of two.

Remark (The recursion). Writing $T = \frac{1}{4}(R_+^2 + R_-^2)$ and $Y = \frac{1}{4}(R_+^2 - R_-^2)$ (Proposition 3.1) is exactly the AM–GM cone construction of [1] applied one level down, to the pair (R_+^2, R_-^2) in place of (x, y) . The entire apparatus — sum, difference, geometric mean, the cone identity $X^2 + Y^2 = T^2$ — reappears unforced, built from the two radii of the position spinor's own epicyclic decomposition.

6. THE HYPERBOLA: A HALF-RAPIDITY BOOST

[4], Proposition 4.1, shows L 's rotation of (X, Y, T) by angle θ is the double cover of a half-angle $SO(2)$ rotation of the shape spinor $(\sqrt{x}, \sqrt{y})$. We show the direct hyperbolic-generator analogue for B_X and the constant-product hyperbola.

Theorem 6.1 (B_X is the double cover of a half-rapidity boost). *Under the B_X -flow $x(\phi) = x_0 e^\phi$, $y(\phi) = y_0 e^{-\phi}$ of [2], the epicycle radii transform as an exact boost at half the rapidity:*

$$(4) \quad R_+(\phi) = R_+(0) \cosh(\phi/2) - R_-(0) \sinh(\phi/2), \quad R_-(\phi) = R_-(0) \cosh(\phi/2) - R_+(\phi) \sinh(\phi/2).$$

Proof. $\sqrt{x(\phi)} = \sqrt{x_0} e^{\phi/2} = \sqrt{x_0} \cosh(\phi/2) + \sqrt{x_0} \sinh(\phi/2)$, and similarly $\sqrt{y(\phi)} = \sqrt{y_0} \cosh(\phi/2) - \sqrt{y_0} \sinh(\phi/2)$; summing and differencing gives (4) directly. Verified symbolically and numerically to machine precision. $\square$

Corollary 6.2 (The hyperbola recurs one level down). $R_+^2 - R_-^2 = 4Y$ holds identically (pure algebra: $(a+b)^2 - (b-a)^2 = 4ab$), and is preserved along the B_X -flow of Theorem 6.1, since Y is exactly what B_X holds fixed. So the epicycle-radius pair (R_+, R_-) itself traces a hyperbola of the identical shape $R_+^2 - R_-^2 = (2\sqrt{Y})^2$ as the orbit moves along the cone's own hyperbola $T^2 - X^2 = Y$ is held fixed at Y .

By [5], the cone's constant-product hyperbola is exactly the Ellis–Bronnikov wormhole embedding $r^2 = b_0^2 + l^2$ with throat $b_0 = \sqrt{Y}$. Corollary 6.2 says the same shape, with throat $2\sqrt{Y}$, governs the epicycle radii themselves as the underlying orbit is varied along that hyperbola — the wormhole identification recurring one level down, exactly as the AM–GM construction itself did in Section 5.

7. SUMMARY TABLE

cone curve	fixed quantity	epicycle reading
row/column (parabola)	x or y	$Y = L_u/\omega$ (Section 4)
spinor-difference (parabola)	$T - Y$	$R_-^2 = 2(T - Y)$ (Prop. 5.1)
spinor-sum (parabola)	$T + Y$	$R_+^2 = 2(T + Y)$ (Prop. 5.1)
constant-product (hyperbola)	Y	$R_+^2 - R_-^2 = 4Y$, boost at $\phi/2$ (Thm. 6.1)

8. SCOPE

We are explicit about what is and is not established. Theorem 2.1, Proposition 3.1, Proposition 5.1, and Theorem 6.1 are exact identities, verified symbolically (and Theorem 6.1 additionally numerically). We do not claim that L and B_X 's actions on the shape spinor are the same operators as any generator of the spinor's own fictitious-time dynamics; [4]'s own Proposition 4.1 and remark already settle the rotation case (same type of generator, different operator), and nothing here revisits that question for B_X — Theorem 6.1 is a statement about how the *shape spinor changes across a family of orbits*, exactly parallel in kind to Proposition 4.1, not a claim about the boost acting on a single orbit's own time evolution. We also do not attempt, here, to connect X, Y (only T is treated in [4]'s perturbed system) to the epicycle language under perturbation; doing so is a natural next step this note does not take.

AI ASSISTANCE DISCLOSURE

Large language model tools (Anthropic's Claude) were used in the derivation, symbolic and numerical cross-checking, and L^AT_EX preparation of this note. This includes: symbolic verification of Theorem 2.1's epicycle decomposition and Proposition 3.1's coordinate dictionary; symbolic and numerical (machine-precision) verification of Theorem 6.1's half-rapidity boost, including the invariance check underlying Corollary 6.2; and identification of the connection to [5]'s wormhole identification. A candidate connection to the Flamm paraboloid of general relativity was investigated and explicitly rejected: the coefficient structure matches only in magnitude, not sign, and is shown to follow from the AM–GM cone's own fixed normalization rather than any shared physical origin; this negative result is recorded here for completeness though the investigation itself is not reproduced. All mathematical claims were independently verified by the author.

REFERENCES

- [1] J. Ebert, *The Cutting Plane: Every Line Through the Multiplication Table Is a Conic Section*, Preprint (2026).
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- [5] J. Ebert, *The Same Hyperbola: The AM–GM Cone's Fermat Curve as the Ellis–Bronnikov Wormhole, and a Chebyshev Classification of Its Algebraic Cousins*, Preprint (2026).

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